Sulzer CP 150-400 — Installation, Operation and Maintenance Manual

Multistage horizontal centrifugal pump | Tag P-101 | Serial SLZ-9174-2016 | Supplied to Bharat Vindhya Petrochemicals Ltd., Unit-2 Utilities Block

1. Scope and application

This manual covers the Sulzer CP 150-400 multistage boiler feed water pump supplied under order BVPL/2015/UT-2/091 and installed as tag P-101 in the Unit-2 Utilities of Bharat Vindhya Petrochemicals Ltd.. A second identical machine is installed as P-102 in duty-standby configuration. The instructions in Section 7 are mandatory: warranty and the stated bearing life are conditional on the replacement criteria of Section 7.4 being observed.

The pump takes suction from the deaerator storage vessel V-201 and discharges through the feed water pre-heater E-301 to boiler B-401. Forced-feed lubrication is supplied by the auxiliary lube oil pump LP-101A mounted on the same skid. Refer to P&ID PID-U2-1010 Rev 3 for the process connections.

2. Technical data

Parameter	Value
Service	Boiler feed water, deaerator to pre-heater
Rated flow	148 m3/h
Rated head	395 m
Rated speed	2965 rpm
Driver	250 kW, 6.6 kV squirrel-cage induction motor
Casing	Radially split, 12% Cr steel
Impeller stages	4
DE bearing	SKF 6316 C3 deep-groove ball bearing
NDE bearing	SKF 7314 BECBM angular-contact ball bearing
Lubrication	Forced feed, ISO VG 46 turbine oil, 18 l/min at 2.1 bar
Suction source	V-201 deaerator storage vessel
Discharge destination	E-301 feed water pre-heater
Mechanical seal	Cartridge, API Plan 11 flush

3. Installation

- Grout the baseplate and confirm flatness within 0.05 mm/m before coupling alignment.
- Align the pump to the driver within 0.05 mm rim and 0.03 mm face, cold, with piping connected. Record the readings in the commissioning file.
- Suction piping from V-201 must fall continuously to the pump suction flange. A pocket in the suction line will cause vapour binding at low deaerator level.
- Do not use the casing to support the discharge pipework. Flange loads must not exceed the values in Appendix C.

4. Operation

Start against a closed discharge valve with the minimum-flow recirculation line open. Continuous operation below 40 m3/h will overheat the casing and shorten bearing life through increased radial thrust. Auto changeover to the standby machine is initiated on low discharge pressure at PI-1015.

5. Lubrication system

Both bearing housings are lubricated by forced feed from LP-101A (Rotodel RGP-18), rated 18 l/min at 2.1 bar with ISO VG 46 turbine oil. The lube oil header pressure is indicated at PI-1016.

Lubrication parameter	Normal	Alarm	Trip
Header pressure (bar)	2.1	1.4 low	1.1 low-low
Oil temperature (degC)	45 - 60	70	80
Oil cleanliness (ISO 4406)	16/14/11	18/16/13	20/18/15
Iron content (ppm)	below 10	20	35

Loss of lube oil header pressure below 1.4 bar removes the oil film from the rolling elements within minutes at rated speed. Sustained operation below this value will destroy the non-drive-end bearing.

6. Condition monitoring

Measured parameter	Location	Alarm	Trip	Instrument
Overall vibration (mm/s RMS)	NDE bearing housing	7.1	9.5	VT-1011
Bearing metal temperature (degC)	NDE and DE housings	85	95	TE-1011A/B
Discharge pressure (barg)	Pump discharge	38 low	35 low-low	PI-1015
Lube oil header pressure (bar)	Bearing oil header	1.4 low	1.1 low-low	PI-1016

Vibration spectra shall be recorded quarterly with a data collector capable of resolving 1x and 2x running speed. A rising 1x component with a 2x sideband at the non-drive end is the characteristic signature of outer-race wear on the angular-contact bearing and precedes measurable temperature rise by several weeks.

7. Maintenance

7.1 Routine schedule

Task	Interval	Reference
Lube oil sample and analysis	Quarterly	Section 5
Vibration survey with spectrum	Quarterly	Section 6
Bearing wear assessment	Quarterly	Section 7.4
Mechanical seal inspection	Annually	Section 7.6
Full overhaul	5 years or on condition	Section 8

7.4 Bearing replacement criteria — MANDATORY

Bearing replacement is mandatory once measured wear reaches 85% of the permissible limit defined in Table 7-2. Wear at or above 85% shall not be carried into the next operating quarter under any circumstances, and shall not be deferred against production requirements. Replacement is to be executed by a competent person following the plant's own bearing replacement procedure.

Wear (% of limit)	Classification	Required action
Below 60	Normal	Continue quarterly monitoring.
60 to 74	Elevated	Increase monitoring to monthly. Plot the trend against the 85% limit.
75 to 84	Pre-replacement	Raise the replacement work order and order the bearing set. Confirm the spare is on site.
85 and above	Replacement mandatory	Replace at the next available opportunity. Do not carry into the next quarter.

Wear is assessed by comparing outer-race shell thickness and load-zone scoring against the reference photographs in Appendix D. Where the wear pattern matches the scoring signature of a previous seizure on the same machine class, treat the reading as pre-replacement irrespective of the measured percentage.

7.6 Bearing arrangement

- Drive end: SKF 6316 C3 deep-groove ball bearing — deep-groove ball bearing, radial load only.
- Non-drive end: SKF 7314 BECBM angular-contact ball bearing — angular contact, carries the residual axial thrust. This is the bearing that fails first.
- Never flame-heat a bearing. Induction heating to a maximum of 110 degC only.

8. Spare parts and process interfaces

Item	Part number	Recommended stock
NDE bearing (angular contact)	SKF 7314 BECBM	1 set
DE bearing (deep groove)	SKF 6316 C3	1 set
Cartridge mechanical seal	SLZ-MS-150-400	1
Bearing housing lip seals	SLZ-LS-314	2 sets
Lube oil pump rotor/idler set	RD-RGP-18-ROT	1

Process interfaces: suction from V-201 (Deaerator Storage Vessel V-201); discharge to E-301 (Feed Water Pre-Heater E-301); lubrication from LP-101A. Refer to P&ID PID-U2-1010 Rev 3.